

A Project report on
“Inventory Management System
For Rifat Enterprise”

KAMANI SCIENCE & PRATAPRAI ART'S COLLAGE AMRELI

Developed By:

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Guided By::

Prof. Paraskumar Thummar

Intern Supervisor

Altaf Naya

Year Of Submission

On

Submitted In partial fulfilment of the requirement

For the award of the degree of Bachelor of Computer Application

(BCA Sem-6)

2025-2026

Saurashrta University-Rajkot



RIFAT ENTERPRISE

— Brick, Cement Block & Interlocking Tile Manufacturing —

● Khodvadri, Gariyadhar,
Bhavnagar - 364505
✉ altafnaya55@gmail.com

Date:

To,

Concerned Authority

Subject: Internship Completion Certificate

This is to certify that Mr. Bukera Inayat has successfully completed 120 hours of internship at Rifat Enterprise from 01/12/2025 to 15/03/2026.

During the internship period, he worked on the development of a Web-Based Inventory Management System for our organization. He demonstrated dedication, technical knowledge, and professionalism while completing tasks related to raw material management, production tracking, sales management, and reporting modules.

His performance during the internship was satisfactory, and we wish him success in his future academic and professional endeavours.

Sincerely,

Authorized Signatory

Rifat Enterprise

✉ altafnaya55@gmail.com

"Quality Bricks, Trusted Service"



RIFAT ENTERPRISE

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• Khodvadri, Gariyadhar,
Bhavnagar - 364505
✉ altafnaya55@gmail.com

Date:

To,

Concerned Authority

Subject: Internship Completion Certificate

This is to certify that Mr. Boricha Jaysukh has successfully completed 120 hours of internship at Rifat Enterprise from 01/12/2025 to 15/03/2026.

During the internship period, he worked on the development of a Web-Based Inventory Management System for our organization. He demonstrated dedication, technical knowledge, and professionalism while completing tasks related to raw material management, production tracking, sales management, and reporting modules.

His performance during the internship was satisfactory, and we wish him success in his future academic and professional endeavours.

Sincerely,

Authorized Signatory

Rifat Enterprise

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"Quality Bricks, Trusted Service"

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Preface

The main objective of this project is to gain practical knowledge and hands-on experience in web development, which is an essential skill for any computer science student. Developing an **Inventory Management System for Rifat Enterprise** allowed us to combine both theoretical concepts and practical implementation, providing a comprehensive learning experience.

During the course of this project, we explored various aspects related to system design, database management, user interface development, and inventory tracking. This process helped us enhance our problem-solving abilities and understand the challenges involved in developing a real-world business application.

Through this project, we learned how to manage raw materials, production data, product stock, and sales information efficiently using a web-based system. The development process also helped us understand how software solutions can improve business operations by replacing manual record-keeping with a structured digital system.

We are pleased to present this project report, which has been carefully organized for easy understanding. Throughout the development process, several software engineering principles and best practices have been applied to ensure that the system is reliable, efficient, and user-friendly. This project has been an important step in strengthening our technical knowledge and preparing us for future opportunities in the field of software development.

Acknowledgment

We, the students of BCA Semester-5 at **kamani Science College**, Amreli, would you like to express our heartfelt Gratitude We sincerely thank **Prof. Paraskumar Thummar** for their invaluable guidance, motivation, and constant support of this project. Their insights and encouragement have played a crucial role in helping us understand The technical and conceptual aspects of the system.

We are also deeply grateful to the head of Department for providing us with the opportunity and inspiration to take this project. A special thanks to all faculty members and administrative staff of our institute for their continued support and resources.

Lastly, we extend our warm thanks to our friends for their help and cooperation , whether directly or indirectly, and to our families for their endless encouragement and support throughout this journey.

Thank to all !

Bukera Inayat

Boricha Jaysukh.

Abstract

The **Rifat Enterprise Inventory Management System** is a web-based platform designed to efficiently manage and monitor the inventory operations of the organization. This system aims to streamline stock tracking, product management, and overall inventory control through a centralized and user-friendly interface.

The project focuses on developing a modern and efficient system where administrators and staff can add, update, and manage product details, monitor stock levels, and maintain accurate records of incoming and outgoing goods. It includes features such as product categorization, real-time stock updates, supplier management, sales tracking, and automated alerts for low inventory levels.

The platform benefits the organization by reducing manual errors, improving operational efficiency, and ensuring better decision-making through accurate data management. With an intuitive layout and secure access controls, the system enhances productivity and helps Rifat Enterprise maintain optimal stock levels while minimizing losses and delays.

Additionally, the system supports report generation and data analysis, enabling management to track inventory trends and improve future planning. Overall, the Inventory Management System serves as a reliable and scalable solution for modern business inventory operations.

Introduction

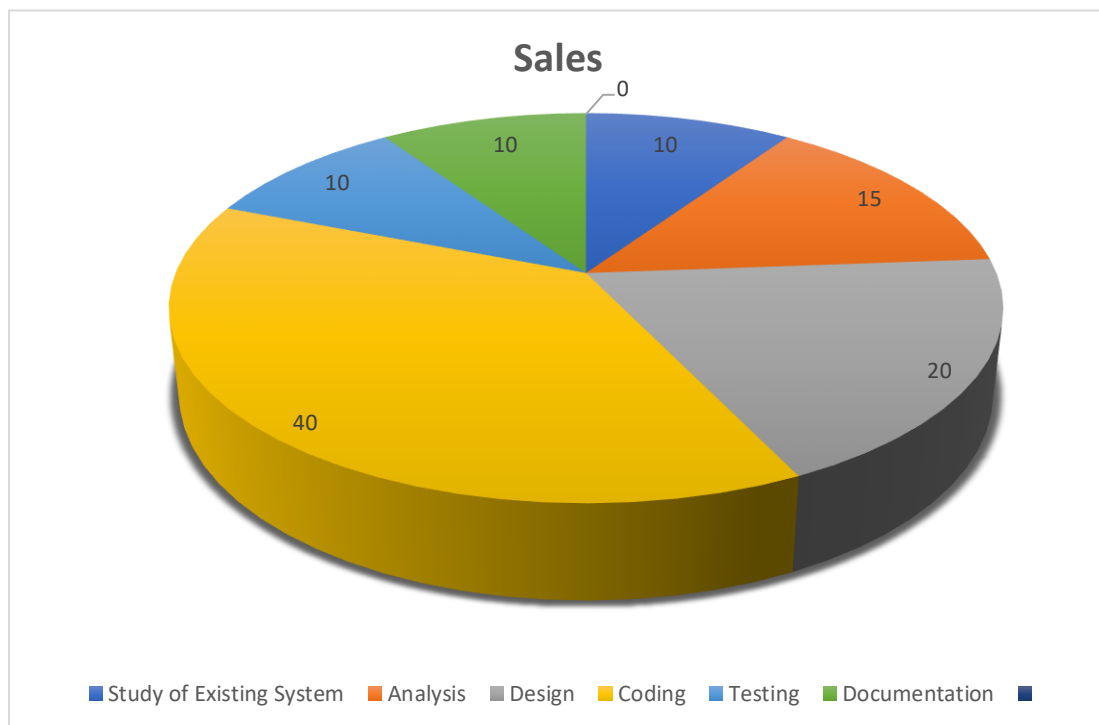
An Inventory Management System for Rifat Enterprise for Blocks is a software-based solution designed to efficiently track and control the business stock of concrete blocks, raw materials, and related supplies.

The system helps the enterprise maintain accurate records of incoming and outgoing blocks, monitor stock levels in real time, and prevent shortage or overstock situations.

By organizing inventory data such as supplier details, quantities in hand, and transaction records, the system supports better planning, reduces manual errors, and improves overall operational efficiency in fulfilling customer orders and managing daily transactions.

The proposed system aims to replace the traditional manual method with a **digital inventory management system** that improves accuracy, efficiency, and reliability. By using this system, the administrator can easily record and monitor all important business activities such as raw material purchases, block production, finished goods stock, and sales transactions.

Time Line



Time Line Chart

1. Study Of Existing System → 10 Days
2. System Analysis → 15 Days
3. System Design → 20 Days
4. Coding → 40 Days
5. Testing → 10 Days
6. Documentation → 10 Days

System Analysis

Rifat Enterprise Inventory Management System – Q&A

Q1. What is the name of your factory?

⇒ Rifat Enterprise.

Q2. Who can use this system?

⇒ Admin only.

Q3. What are the main benefits of the website?

⇒ The system allows the admin to manage raw materials, products, and sales efficiently.

Q4. What is the problem definition statement and why does Rifat Enterprise need an inventory management system?

⇒ Rifat Enterprise currently faces difficulties in manually managing raw materials, production records, stock levels, and sales. Manual record keeping often leads to errors, stock mismatches, and poor decision making. Therefore, an inventory management system is required to automate stock tracking and manage production and sales more efficiently.

Q5. What are the functional and non functional requirements?

⇒ Functional requirements include managing raw materials, production, finished goods, sales, suppliers, customers, and generating analysis reports.

Non functional requirements include system security through user authentication, fast response time, data accuracy, reliability, and an easy to use interface.

Q6. What are the technical, economic, and operational feasibilities of developing the system?

⇒ The system is technically feasible because it is developed using PHP, MySQL, HTML, CSS, and JavaScript, which are widely available and easy to implement technologies.

Q7. Which type of system architecture is followed?

⇒ Three tier architecture consisting of User Interface (UI), Application Logic, and Database layers.

Q8. How is stock updated in the system?

⇒ Stock is automatically updated during production entries and sales entries.

Q9. Which type of blocks are produced?

⇒ Two types of blocks are produced:

1. Bricks
2. Cement Blocks

Q10. How is system security maintained?

⇒ System security is maintained through user authentication, role based access control, and session management.

Q11. What is raw material management in your system?

⇒ Raw material management allows the admin to **add, update, and monitor the stock levels of materials such as cement, sand, aggregate, and red sand used in production.**

Q12. What is production management in the system?

⇒ Production management records the **daily production of bricks and cement blocks and automatically updates raw material consumption and finished product stock.**

Q13. How does the system help in sales management?

⇒ The system records **product sales, customer information, quantity sold, and total amount**, which helps in tracking revenue and stock levels.

Q14. What are the limitations of your system?

⇒ The system currently supports **only admin access and basic reporting. Advanced analytics and multi-user roles can be added in future versions.**

Q15. What testing methods were used during development?

⇒ The system was tested using **unit testing, integration testing, and system testing to ensure accuracy and reliability.**

Q16. What is the role of suppliers in the system?

⇒ Suppliers provide raw materials, and their details are stored in the system to **track material purchases and supplier transactions.**

Profile Page

PROJECT DETAILS	
Project Name	Inventory Management System for Rifat Enterprise
Project Details	System used for managing raw materials, production, sales, customers, suppliers, and easy data analysis.
Front-end Tools	HTML, JavaScript, CSS
Back-end Tools	PHP, MySQL
Operating System	Windows 10/11, Linux, MacOS
Completion Time	3.5 Months
TEAM INFORMATION	
Developed By	Boricha Jaysukh EN. No: 23SI002UG051780 Bukera Inayat EN. No: 23SI002UG05178
Guided By	Prof. ParasKumar Thummar
Submitted To	Saurashtra University, Rajkot

Saurashtra University, Rajkot

Hardware & Software Requirement

Hardware Requirement

- 512+MB of Ram
- Multimedia keypad and mouse
- GHz processor
- Monitor

Software Requirement

- Operating System: Microsoft Window 10
- Front-end Tool: HTML, JS,CSS
- Back-end Tool: PHP, Mysql
- Documentation Tool: MS Visio

Problem Definition

Rifat Enterprise currently manages its inventory using manual methods such as paper records and basic spreadsheets. This traditional system creates several operational challenges that affect the efficiency, accuracy, and productivity of the business.

Due to the absence of a computerized inventory system, the organization faces difficulties in tracking stock levels, updating product information, monitoring supplier details, and generating reports. Manual data entry increases the chances of human error, duplication of records, misplaced files, and incorrect stock calculations.

The existing system also makes it hard to:

- Maintain real-time stock updates
- Track sales and purchase records efficiently
- Identify low-stock or out-of-stock items
- Generate accurate billing and invoices
- Prepare monthly or yearly inventory reports
- Maintain proper supplier and customer records

As the business grows, handling inventory manually becomes time-consuming and inefficient. Delays in stock updates may lead to overstocking or stock shortages, resulting in financial loss and poor customer satisfaction.

Therefore, there is a need to develop a computerized **Inventory Management System (IMS)** for Rifat Enterprise. The proposed system will automate inventory tracking, manage product details, maintain stock records, generate bills and reports, and improve overall operational efficiency.

The system aims to:

- Reduce manual errors
- Improve data accuracy
- Provide real-time inventory updates
- Enhance decision-making through reports
- Save time and operational costs

Feasibility Study

1). Technical feasibility:

The proposed system can be developed using standard web technologies such as:

Front-end: Html, Css, JavaScript

Back-end : PHP,

Database : MySQL

Hosting : XAMPP (for local)

The Required Software and tools are easily available and simple to use

2). Operational feasibility:

The System is user-Friendly and easy to operate by staff and admin.

It will reduce manual record keeping and paperwork.

The admin can easily manage raw Materials, Production, and Sales.

The System helps in tracking Stock, Revenue and Profit efficiently.

3). Economic feasibility:

Development costs is low because open source technologies are used.

Hosting and Maintenance Costs are Minimal.

The System Reduce Financial Losses created by Stock Mis-Management

Record Review

Rifat Enterprise

BUILDING MATERIALS SUPPLIER

Khodvadri, Gariyadhar, Bhavnagar, Gujarat, India
altafnaya55@gmail.com

INVOICE

INV-2026-00006

✓ PAID

INVOICE DATE

08 Mar 2026

DUE DATE

07 Apr 2026

REFERENCE NO.

INV-2026-00006

SALE ID

#00006

BILL TO

jatin

Customer

FROM

Rifat Enterprise

Rajkot, Gujarat, India

DESCRIPTION	QTY	UNIT PRICE	AMOUNT
Burnt Clay Bricks Building material — Rifat Enterprise	200 units	₹50.00	₹10,000.00

Subtotal ₹10,000.00

Tax / GST —

Total Due ₹10,000.00

AUTHORISED SIGNATORY

Rifat Enterprise

Generated: 09 Mar 2026, 02:10

Thank you for your business!

INV-2026-00006

Objective Of Project

Welcome to the Inventory Management System developed for **Rifat Enterprise**, a business involved in the manufacturing and trading of bricks, cement blocks, and interlocking tiles. This system is designed to improve the efficiency of managing raw materials, production processes, and sales activities.

The proposed system is a **web-based application** that helps the organization maintain accurate records of raw materials, production details, product stock, and customer sales. It replaces the traditional manual record-keeping system and provides a faster, more reliable way to manage business operations.

The platform allows administrators and staff to easily monitor inventory levels, record production activities, manage sales transactions, and generate reports. This system improves transparency, reduces human errors, and supports better decision-making for the business.

Benefits of the system :

1. Efficient Inventory Management:

The system allows administrators to easily add, update, and track raw materials and finished products, ensuring accurate stock records.

2. Production Tracking:

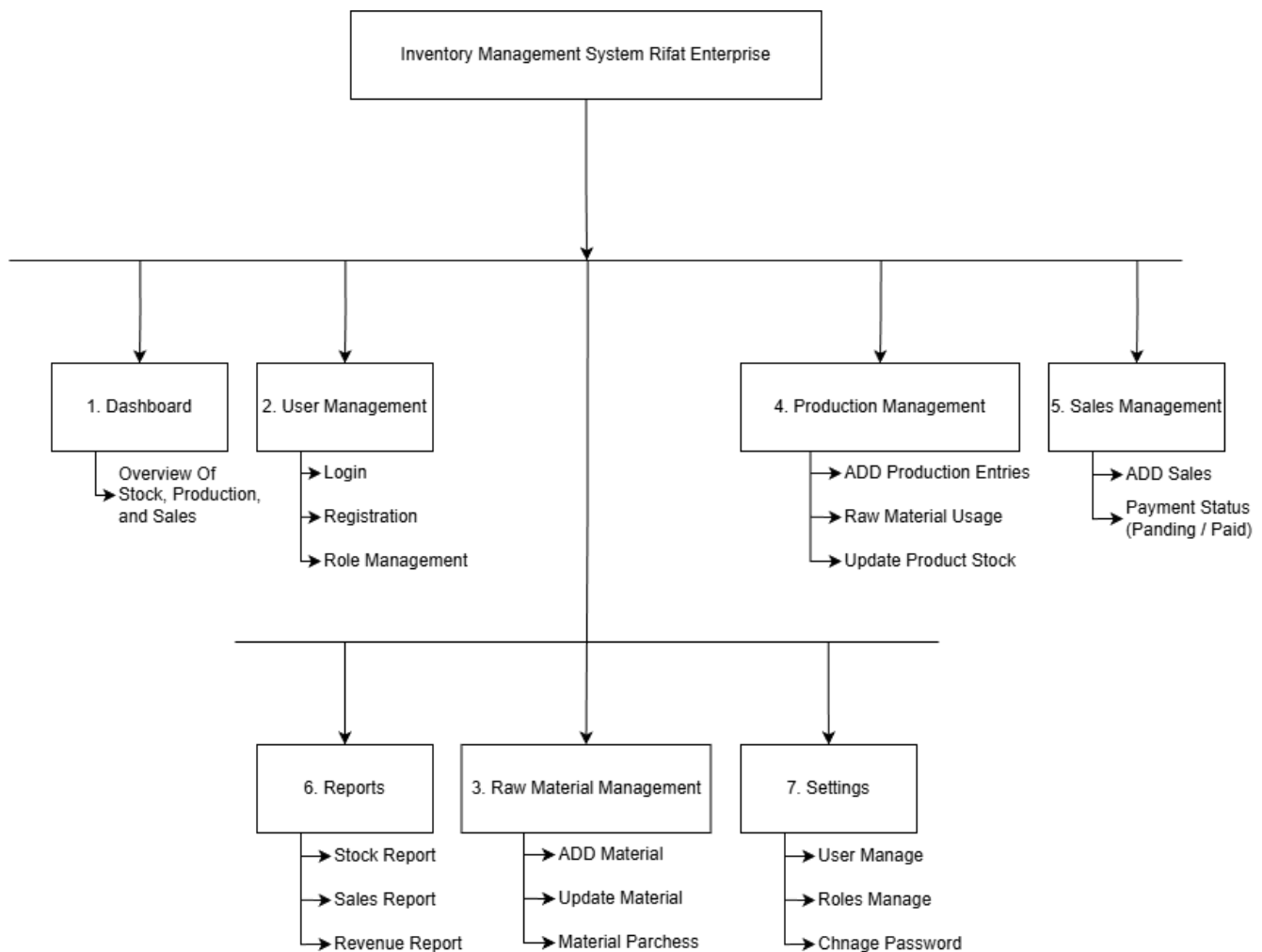
The platform records daily production activities and automatically updates product stock based on raw material usage.

3. Sales Management:

Staff can manage sales transactions details, and generate reports, helping the business monitor revenue and profit.

Project Structure Diagram

1. Diagram :-



Data Dictionary

material_purchases

Column	Type	Null	Default	Links to	Comments	Media type
id (Primary)	int(11)	No				
material_id	int(11)	No		raw_material -> id		
supplier_id	int(11)	No		suppliers -> id		
purchase_date	date	No				
quantity	int(11)	No				
rate	int(11)	No				
total_amount	int(11)	No				
payment_status	int(11)	No				
created_at	datetime	No	current_timestamp()			

production_item

Column	Type	Null	Default	Links to	Comments	Media type
id (Primary)	int(11)	No				
name	varchar(20)	No				
min_stock	int(11)	No				
current_stock	int(11)	No				
price_per_unit	int(11)	No				

production_entries

Column	Type	Null	Default	Links to	Comments	Media type
id (Primary)	int(11)	No				
item_id	int(11)	No		production_item -> id		
new_stock	int(11)	No				
Available	int(11)	No				
aggregate_used	int(11)	No				
sand_used	int(11)	No				
cement_used	int(11)	No				
red_sand_used	int(11)	No				
status	varchar(20)	No				
entry_date	date	No	current_timestamp()			

raw_material

Column	Type	Null	Default	Links to	Comments	Media type
id (Primary)	int(11)	No				
material_name	enum('cement', 'sand', 'Aggregate', 'red sand')	No				
unit	enum('bag', 'kg', 'ton', '')	No				
current_stock	int(100)	No				
min_stock	int(100)	No				
created_at	date	No	current_timestamp()			

roles

Column	Type	Null	Default	Links to	Comments	Media type
id (<i>Primary</i>)	int(11)	No				
role_name	varchar(30)	No				
roles	text	No				
permissions	text	No				

sales

Column	Type	Null	Default	Links to	Comments	Media type
id (<i>Primary</i>)	int(11)	No				
customer	varchar(100)	No				
product_id	int(11)	No		production_item -> id		
qty	int(11)	No				
amount	int(11)	No				
status	enum('paid', 'pending')	No				
date	date	No	current_timestamp()			

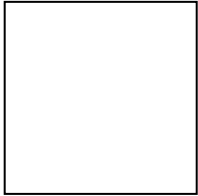
suppliers

Column	Type	Null	Default	Links to	Comments	Media type
id (<i>Primary</i>)	int(11)	No				
name	varchar(30)	No				
phone	text	No				
address	text	No				
created_at	datetime	No	current_timestamp()			

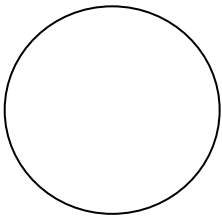
users

Column	Type	Null	Default	Links to	Comments	Media type
id (<i>Primary</i>)	int(11)	No				
username	varchar(30)	No				
email	text	No				
password	varchar(18)	No				
role_id	int(11)	No		roles -> id		

Data Flow Diagram



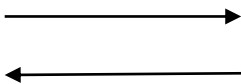
ENTITY



PROCESS

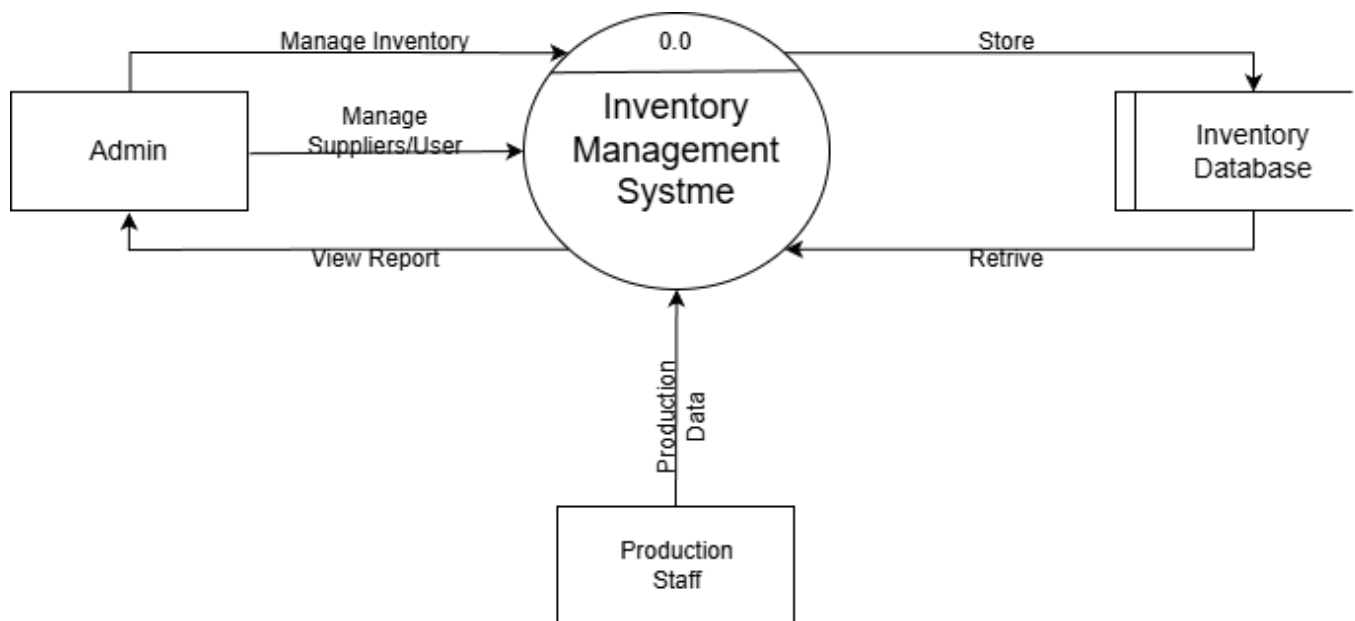


DATA STORE

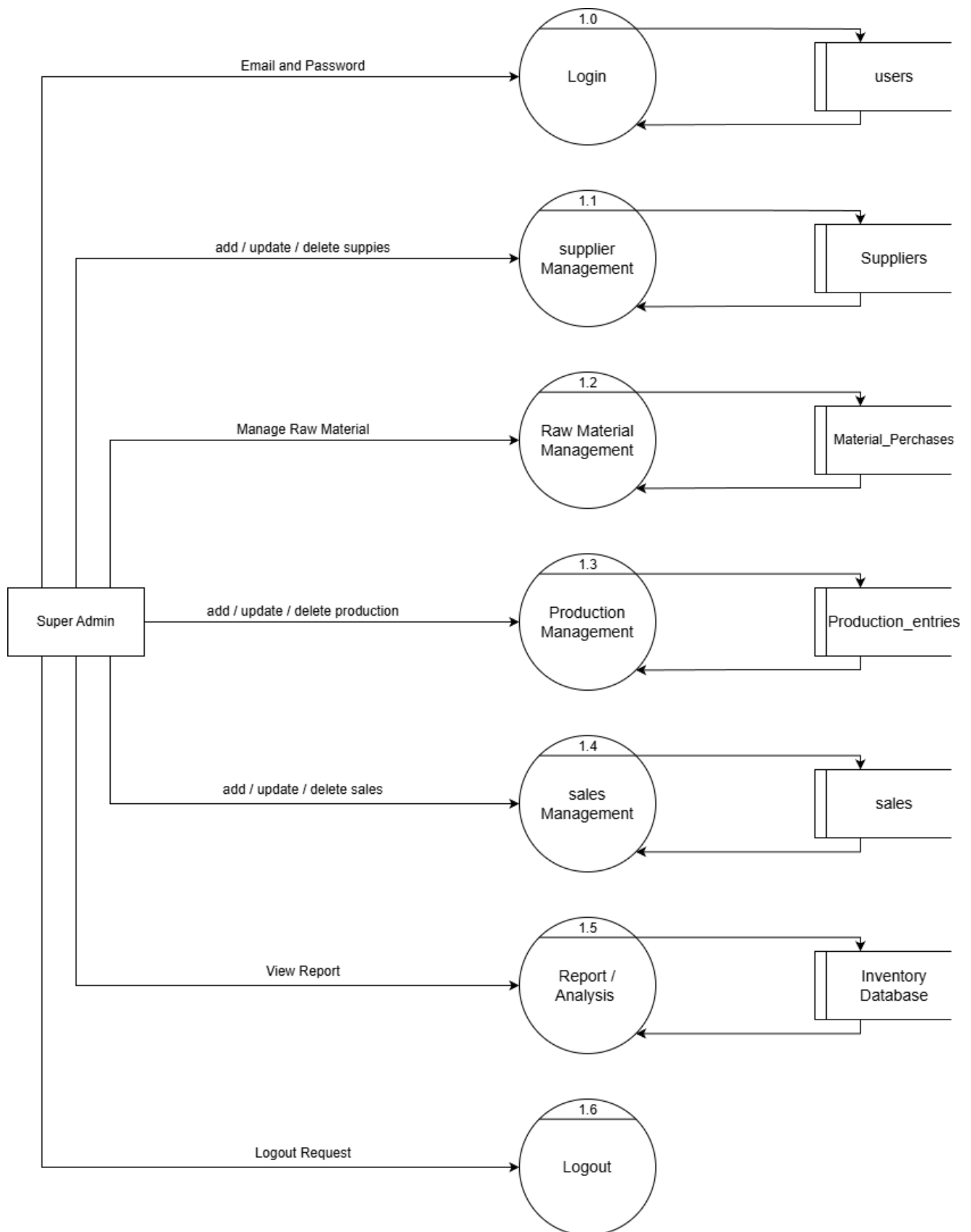


FLOW LINES

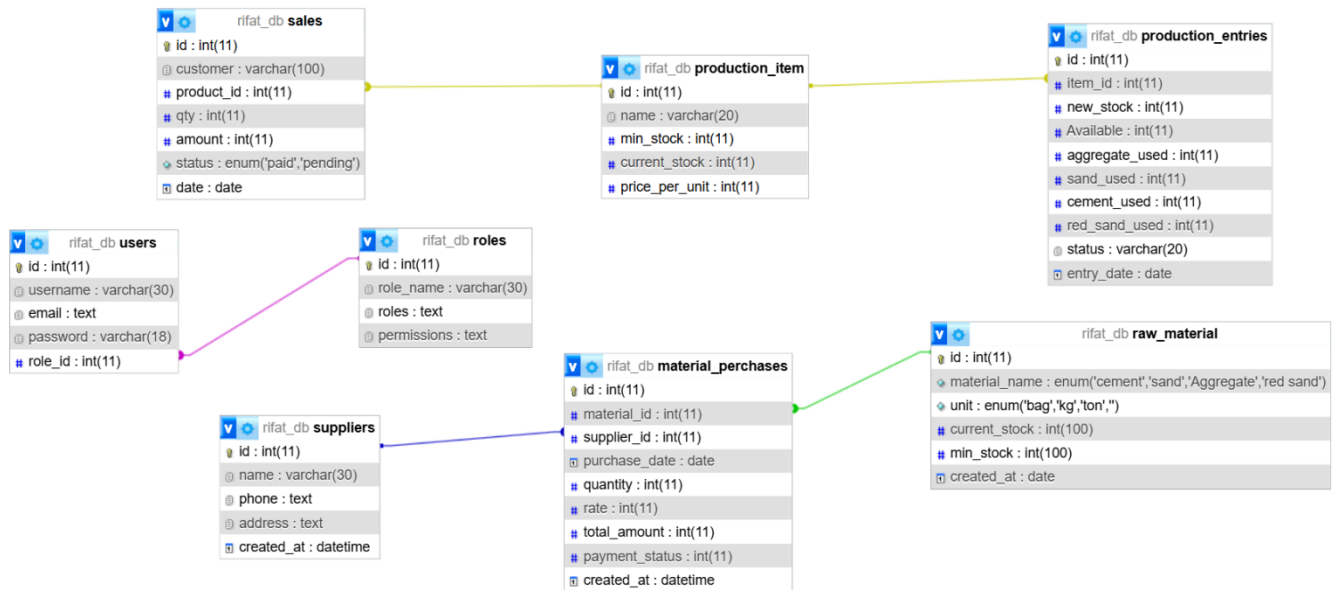
❖ Context Level DFD for Rifat Enterprise IMS(0 level):



❖ First Level DFD for Rifat Enterprise IMS (1 level):



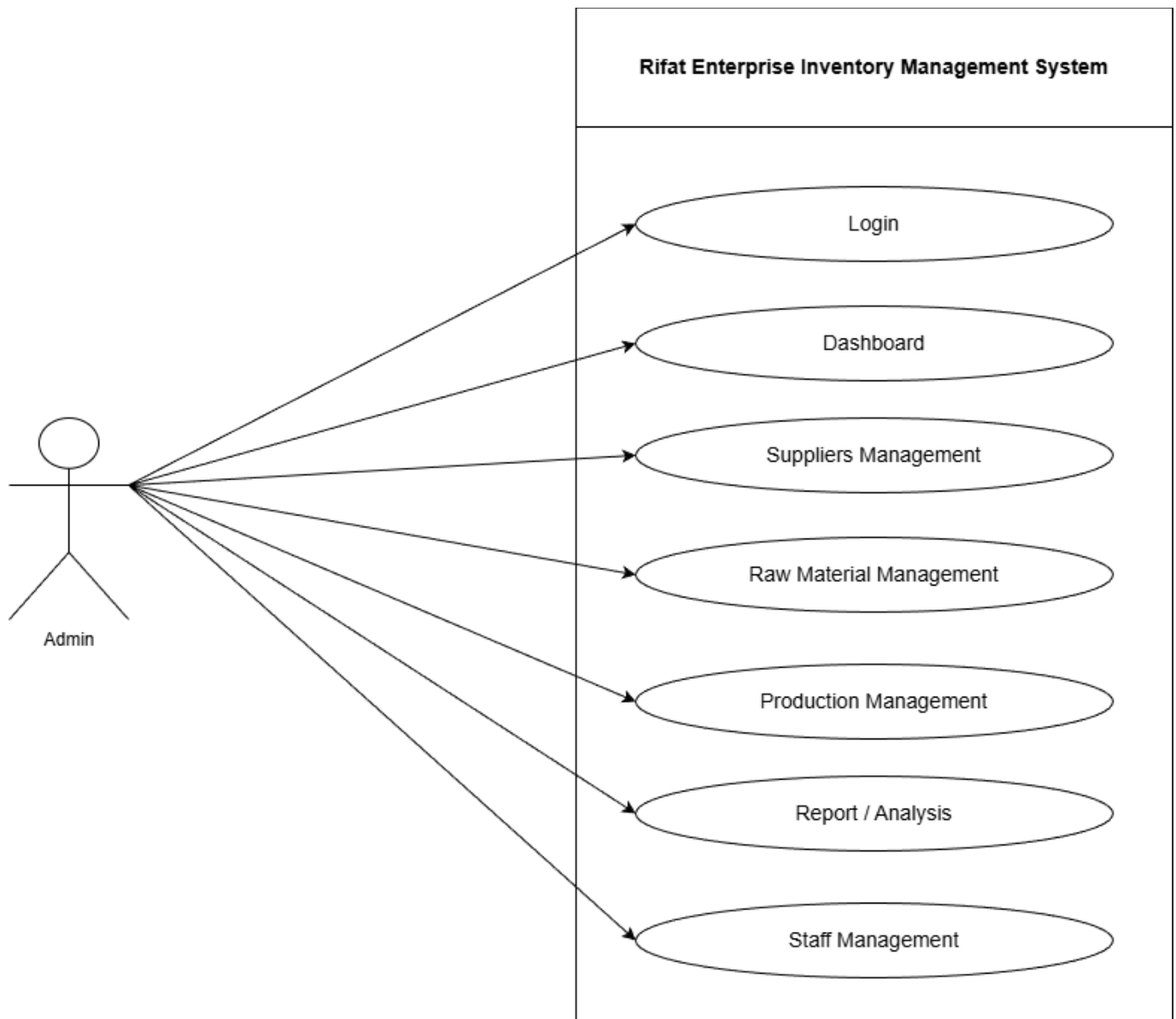
ER Diagram



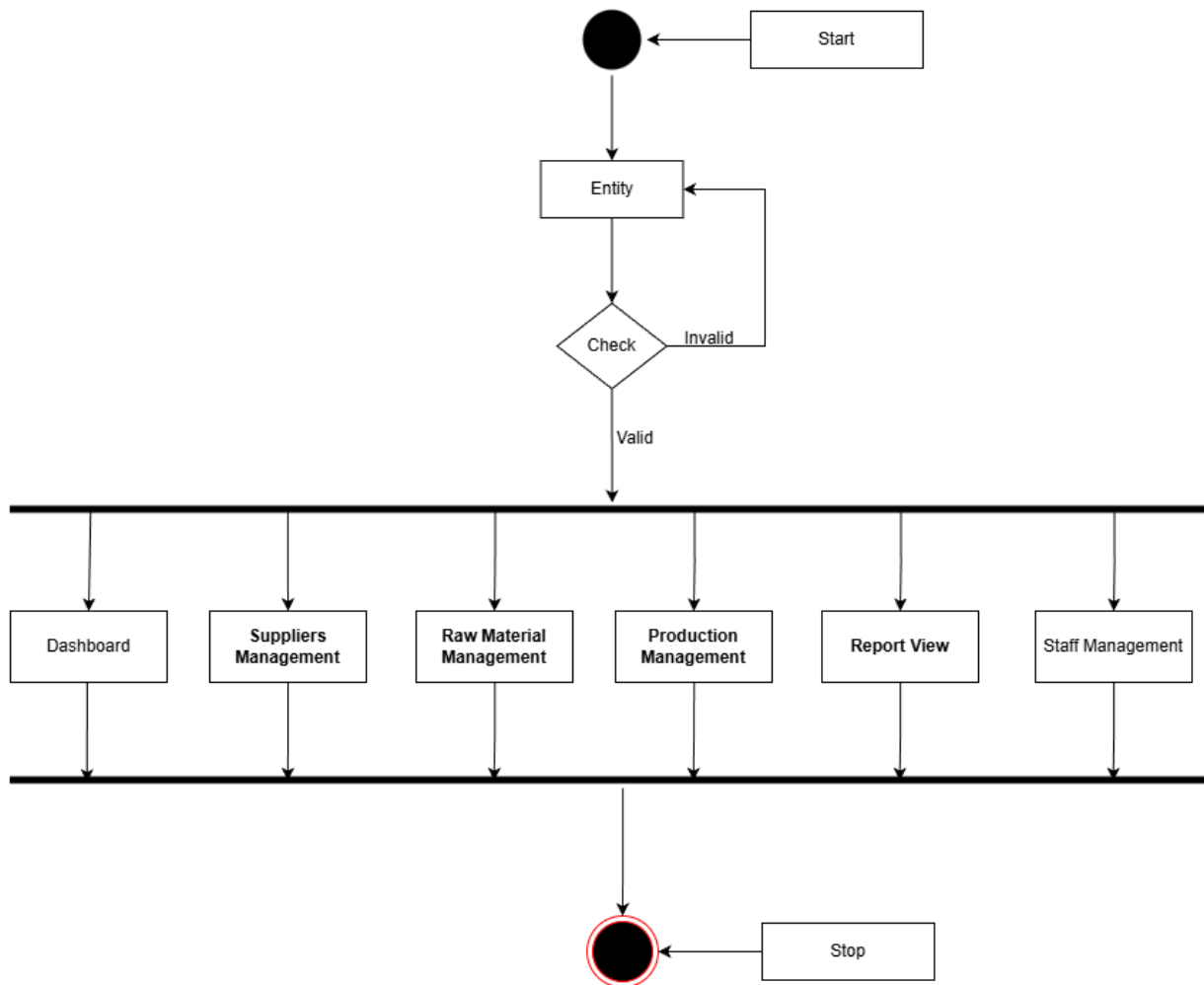
UML Diagram

UML:- A system diagram is a way to visualize systems and software using unified modelling language.

1. Use Case Diagram



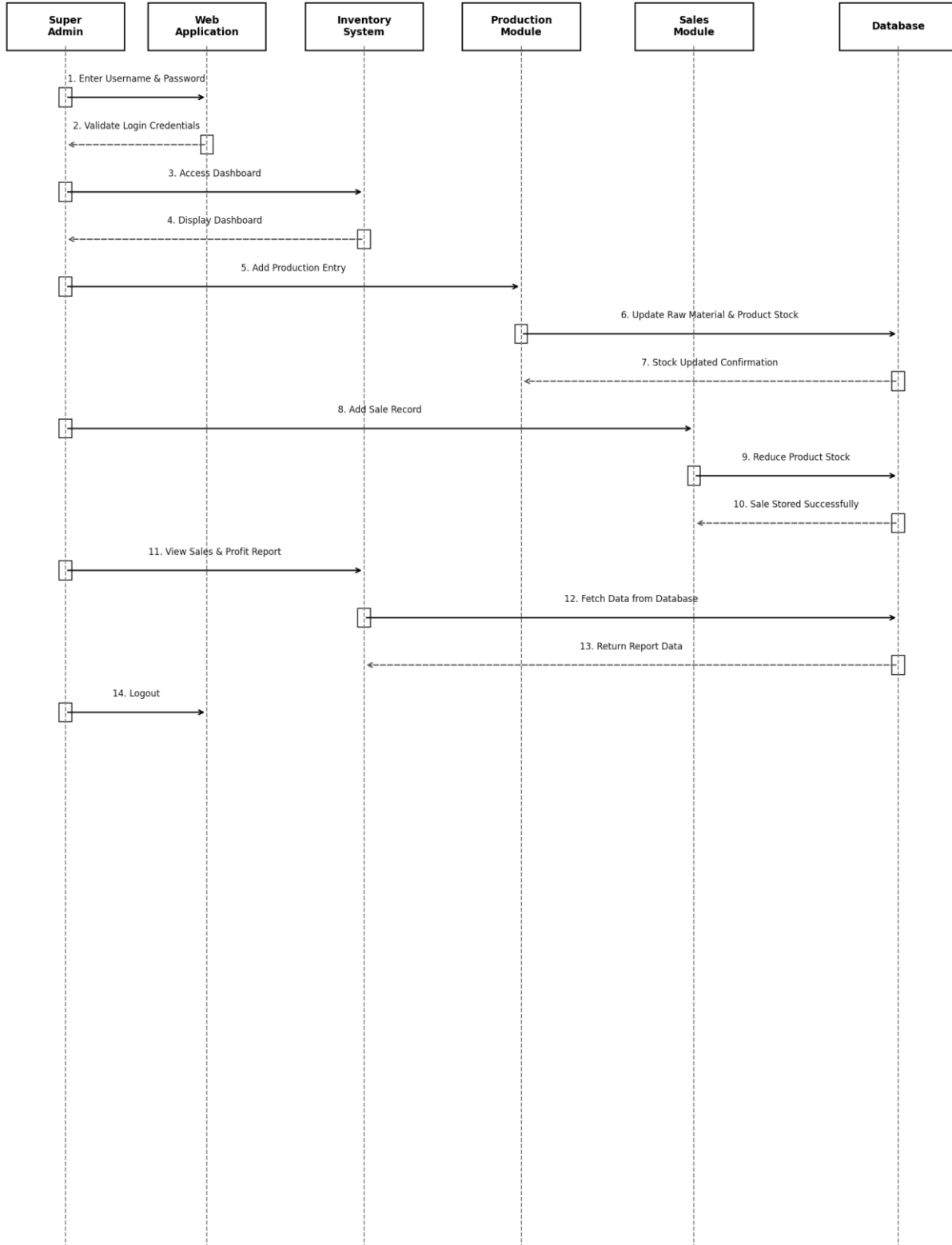
2. Activity Diagram



Sequence Diagram

SEQUENCE DIAGRAM:

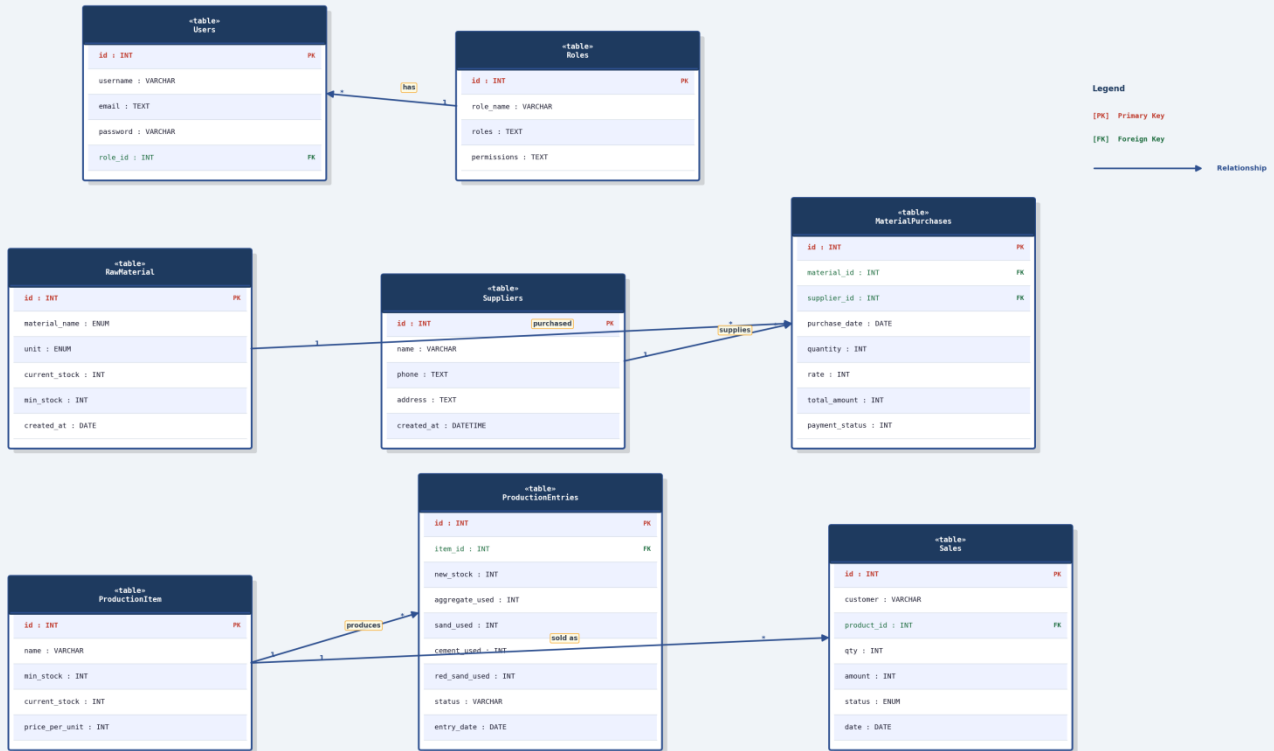
Inventory Management System - Rifat Enterprise



Class Diagram

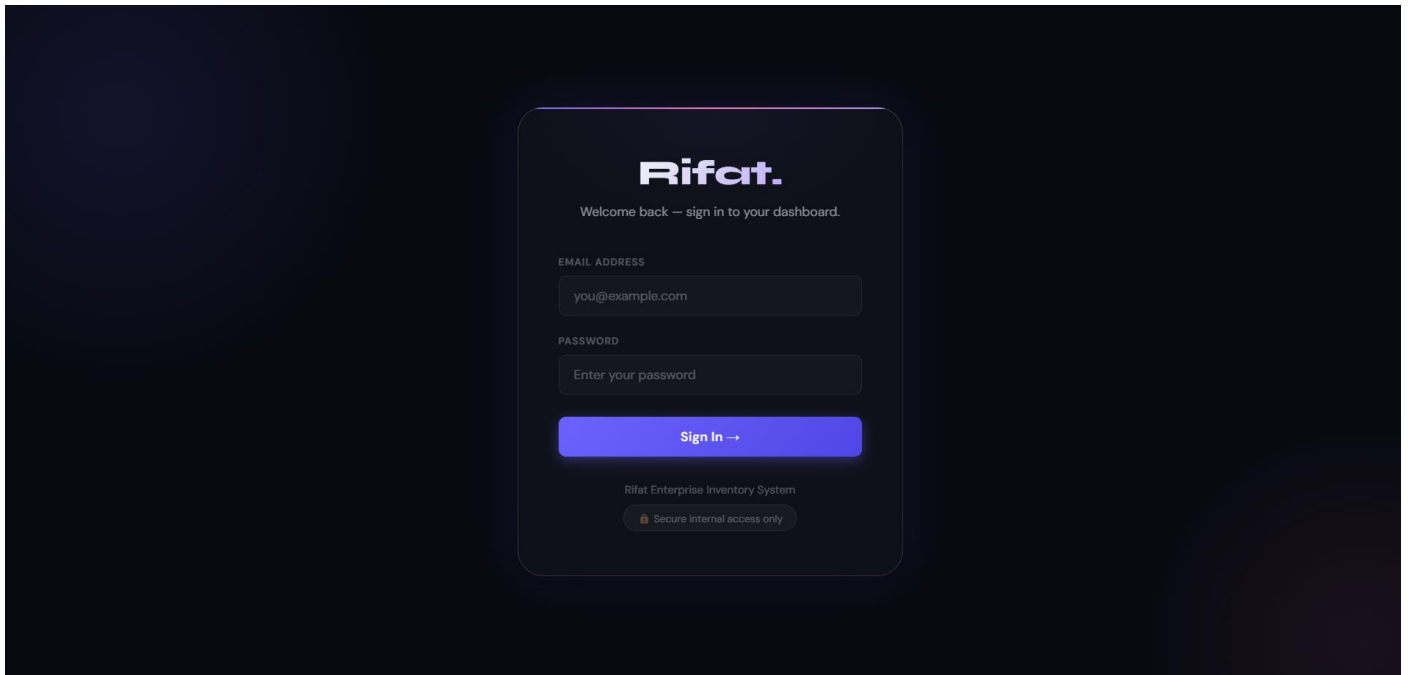
Rifat Enterprise – Inventory Management System

Class / Entity Relationship Diagram



Application Screenshot

Login :-



Dashboard :-

Rifat
Enterprise

Dashboard

Production Management

Raw Material Inventory

Sales Management

Suppliers

Reports

Settings

Inventory Dashboard

altaf

Total Inventory Items

2309

Raw Materials + Finished goods

Product Types

2

Active product lines

Low / Critical Stock

All levels healthy

Monthly Revenue

₹10.00 K

200 units sold this month

Pending Payments

₹1.00 L

Outstanding receivables

Burnt Clay Bricks

Sale

-200

Today

Cement Blocks

Sale

-150

16 days ago

Burnt Clay Bricks

Sale

-120

16 days ago

Cement Blocks

Sale

-1

16 days ago

Cement Blocks

Sale

-100

16 days ago

Cement Blocks

Production

+120

17 days ago

Burnt Clay Bricks

Production

+1200

17 days ago

Inventory Alerts

₹100,005 in pending payments outstanding

Production

Raw Materials

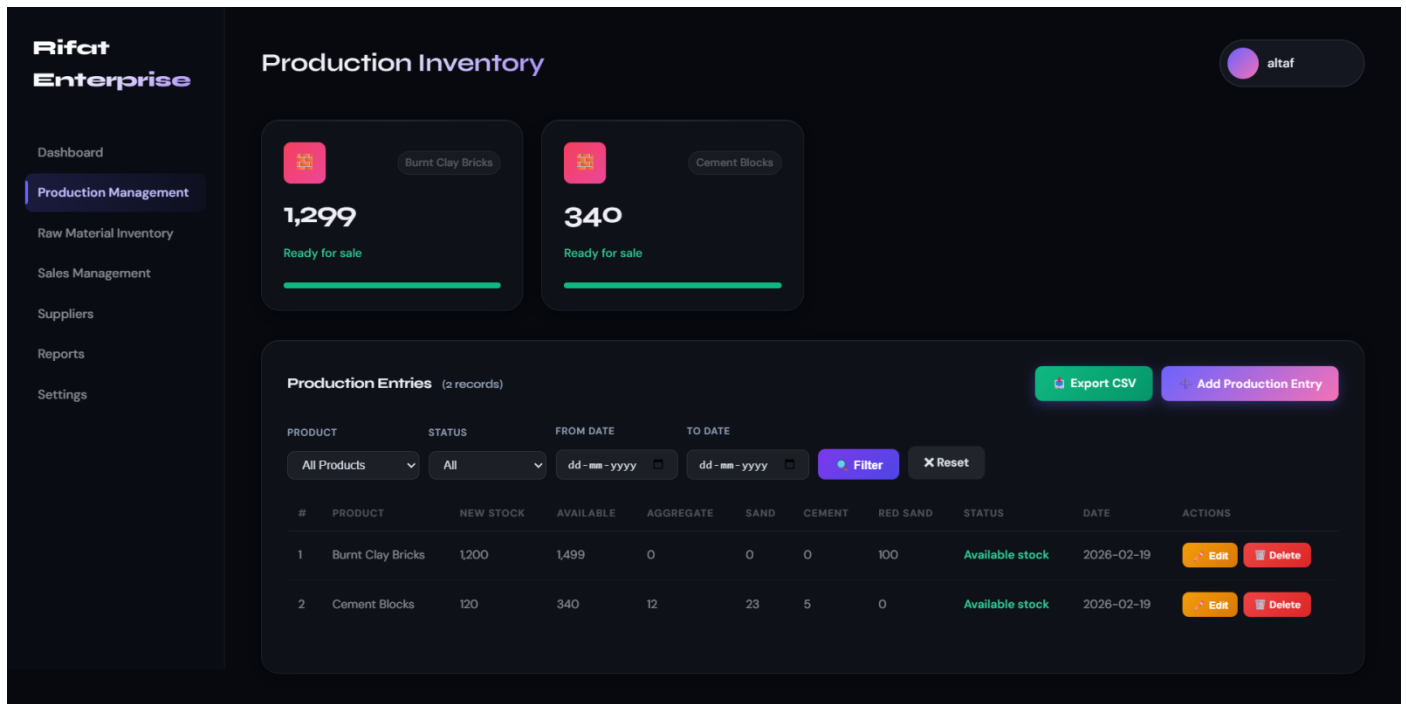
Sales

Suppliers

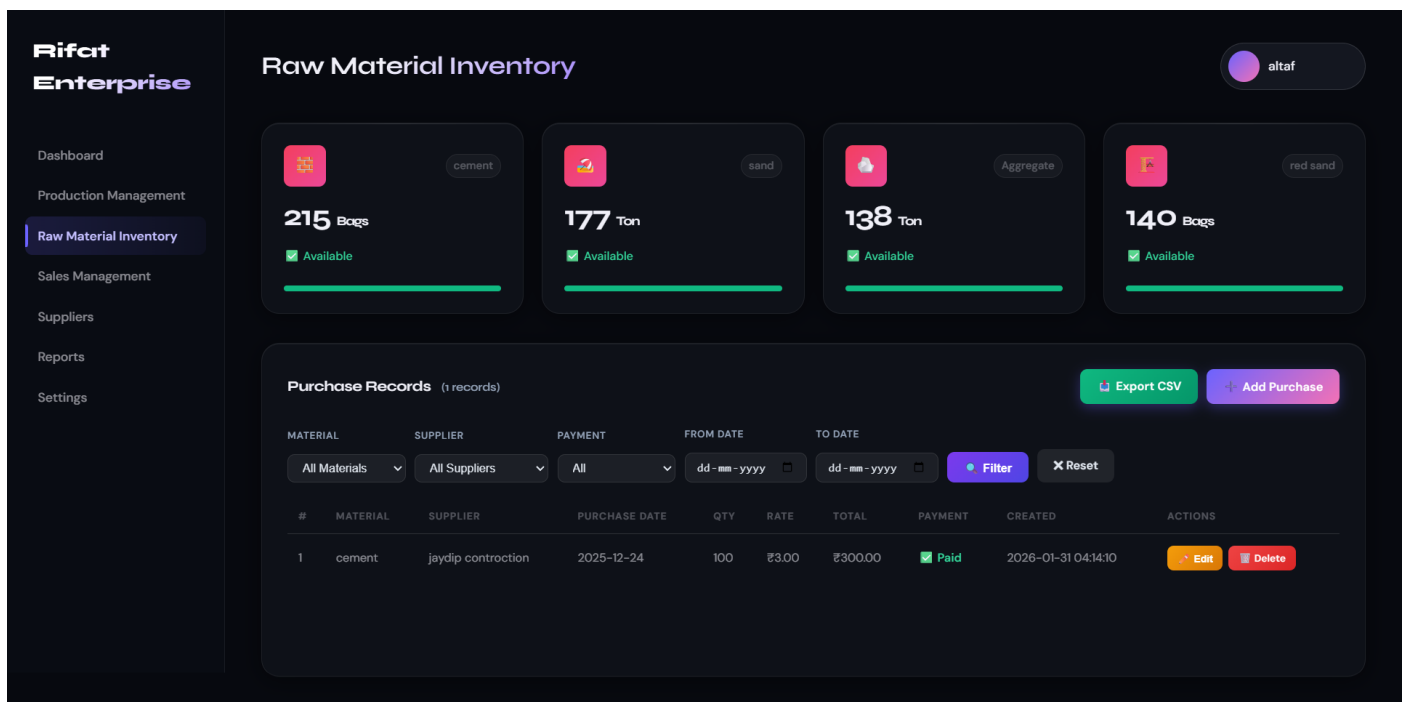
ITEM	CATEGORY	AVAILABLE	REORDER LEVEL	STATUS
Cement	Raw Material	215 bag	50 bag	In Stock
Sand	Raw Material	177 ton	20 ton	In Stock
Aggregate	Raw Material	138 ton	40 ton	In Stock
Red Sand	Raw Material	140 ton	50 ton	In Stock
Burnt Clay Bricks	Finished Goods	1299 units	250 units	In Stock
Cement Blocks	Finished Goods	340 units	120 units	In Stock

30

Production Management :-



Raw Material Management :-



Sales Management :-

Rifat Enterprise

Dashboard

Production Management

Raw Material Inventory

Sales Management

Suppliers

Reports

Settings

Sales Management

Total Sales (Month)

₹10.00 Thousand

Current month

Units Sold

200

Bricks / Blocks / Tiles

Pending Payments

₹1.00 Lakh

Outstanding

Sales Records (6 records)

Export CSVNew Sale

CUSTOMER

PRODUCT

STATUS

FROM DATE

Search name...

All Products

All

dd-mm-yyyy

TO DATE

dd-mm-yyyy

Filter

Reset

#	CUSTOMER	PRODUCT	QTY	AMOUNT	STATUS	DATE	ACTIONS
6	jatin	Burnt Clay Bricks	200	₹10,000.00	Paid	2026-03-08	Mark PendingEdit InvoiceDelete

Sales Alerts

₹100000 Pending payment from jatin

Burnt Clay Bricks stock reducing fast

High demand for Cement Blocks

Stock Levels

Burnt Clay Bricks1,299 units

Cement Blocks340 units

Sales Summary

PRODUCT	TOTAL SOLD (UNITS)	REVENUE	STOCK IMPACT
Burnt Clay Bricks	320	₹10.60 Thousand	Low
Cement Blocks	261	₹1.03 Lakh	High

Suppliers Management :-

Rifat Enterprise

Dashboard

Production Management

Raw Material Inventory

Sales Management

Suppliers

Reports

Settings

Suppliers Management

Total Suppliers

2

Registered vendors

Supplier List (2 records)

Export CSVAdd Supplier

SEARCH

Name, phone, or location...

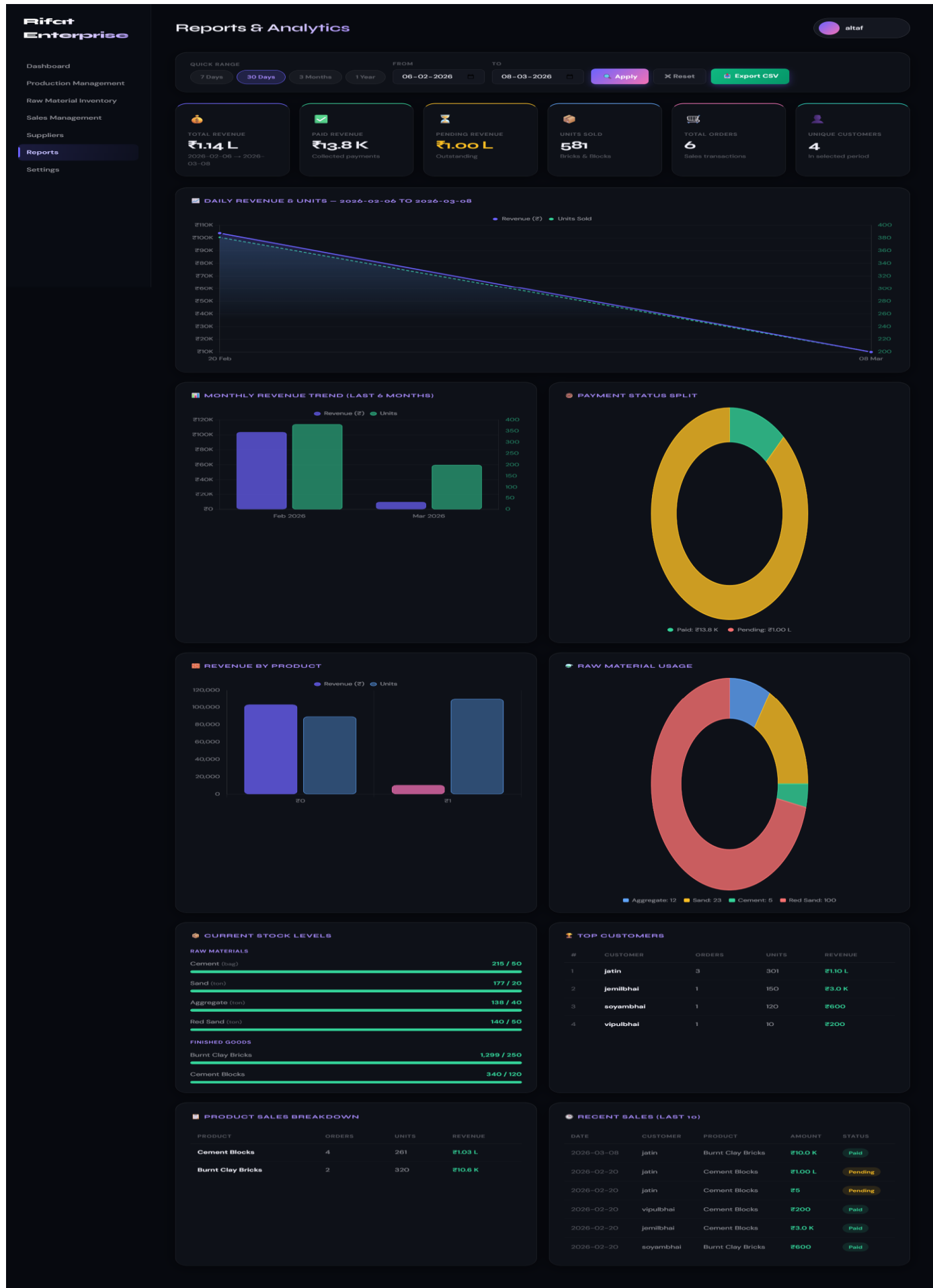
Search

Reset

#	NAME	CONTACT	LOCATION	ACTIONS
1	ambuja cement	6356857545	gariyadhar	EditDelete
2	jaydip controction	8140401738	khodvadri	EditDelete

32

Sales Management :-



Settings :-

Rifcat
Enterprise

Dashboard

Production Management

Raw Material Inventory

Sales Management

Suppliers

Reports

Settings

Settings

Overview

Manage Users

Roles & Permissions

Change Password

Your Account

ALTAF

altafnaya55@gmail.com

super admin

2

Total Users

2

Roles Defined

7

System Pages

Manage Users

Add, edit, or remove system users. Assign roles to control access. Currently **2 user(s)** registered.

Go to Users

Roles & Permissions

Create roles and fine-tune which pages each role can access. Currently **2 role(s)** defined.

Go to Roles

Change Password

Update your own login password. We recommend using a strong, unique password.

Change Password

34

Sales Management :-

Rifcat
Enterprise

Dashboard

Production Management

Raw Material Inventory

Sales Management

Suppliers

Reports

Settings

My Profile



altaf

Account holder

Username

altaf



altafnaya55@gmail.com

Registered email

Email

altafnaya55@gmail.com



Super admin

Access level

Role

Super admin

Account Details

USERNAME	altaf
EMAIL	altafnaya55@gmail.com
ROLE	Super admin

Permissions

☒ Dashboard

☒ Production_management

☒ Raw_material_management

☒ Sales_management

☒ Suppliers

☒ Reports

☒ Settings

Code Screenshot

User Login/Register page :-

```
2  <html lang="en">
10 <body>
12   <div class="login-card">
17   </div>
18   <p class="welcome">Welcome back – sign in to your dashboard.</p>
19
20   <!-- Error message (shown when login fails) -->
21   <?php if (isset($_GET['error'])): ?>
22     <div class="alert-error">
23       <?php
24         $errs = [
25           'invalid' => 'Invalid email or password. Please try again.',
26           'empty'   => 'Please fill in all fields.',
27           'access'  => 'You do not have access to this page.',
28         ];
29         echo htmlspecialchars($_GET['error']) ?? 'Login failed. Please try again.';
30       ?>
31     </div>
32   <?php endif; ?>
33
34   <form action="login.php" method="POST" autocomplete="off">
35
36     <div class="form-group">
37       <label for="email">Email Address</label>
38       <input
39         type="email"
40         id="email"
41         name="email"
42         placeholder="you@example.com"
43         value="<?php echo isset($_GET['email']) ? htmlspecialchars($_GET['email']) : ''; ?>"
44         required
45         autofocus
46       >
47     </div>
48
49     <div class="form-group">
50       <label for="password">Password</label>
51       <input
52         type="password"
53         id="password"
54         name="password"
55         placeholder="Enter your password"
56         required
57       >
58     </div>
59   </form>
```



Home page :-

```
189 <body>
190 <div class="container">
191
192 <?php include "sidebar.php"?>
193
194 <main class="main">
195
196 <div class="header">
197 <h1>Inventory Dashboard</h1>
198 <div class="profile" onclick="window.location.href='dashboard/profile.php'" style="cursor:pointer;">
199 <div class="avatar"></div>
200 <span><?php echo $_SESSION['username'];?></span>
201 </div>
202 </div>
203
204 <section class="stats-grid">
205
206 <div class="stat-card">
207 <div class="stat-top">
208 <div class="stat-icon used">📦</div>
209 <div>Total Inventory Items</div>
210 </div>
211 <div class="stat-number"><?php echo $total_inventory;?></div>
212 <div class="stat-label">Raw Materials + Finished goods</div>
213 </div>
214
215 <div class="stat-card">
216 <div class="stat-top">
217 <div class="stat-icon recent">📦</div>
218 <div>Product Types</div>
219 </div>
220 <div class="stat-number"><?php echo $product_types;?></div>
221 <div class="stat-label">Active product lines</div>
222 </div>
223
224 <div class="stat-card">
225 <div class="stat-top">
226 <div class="stat-icon stats">⬆️</div>
227 <div>Low / Critical Stock</div>
228 </div>
229 <?php if ($low_stock_count > 0) {?>
230 <div class="stat-number" style="color: #f47b7b;"><?php echo $low_stock_count;?></div>
231 <div class="stat-label">Items need attention</div>
232 <?php } else {?>
233 <div class="stat-number" style="color: #28a745;"><?php echo $low_stock_count;?></div>
234 <div class="stat-label">All levels healthy</div>
235 <?php }?>
236 </div>
237
238 <div class="stat-card">
239 <div class="stat-top">
240 <div class="stat-icon used">💰</div>
241 <div>Monthly Revenue</div>
242 </div>
243 <div class="stat-number"><?php echo convert_money($month_revenue);?></div>
244 <div class="stat-label"><?php echo $month_units;?> units sold this month</div>
245 </div>
246
247 <div class="stat-card">
248 <div class="stat-top">
249 <div class="stat-icon stats">📄</div>
250 <div>Pending Payments</div>
251 </div>
252 <?php if ($pending_total > 0) {?>
253 <div class="stat-number" style="color: #ffc107;"><?php echo convert_money($pending_total);?></div>
254 <?php } else {?>
255 <div class="stat-number" style="color: #28a745;"><?php echo convert_money($pending_total);?></div>
256 </div>
257
258 $query = mysql_query($conn, "SELECT SUM(amount) FROM sales WHERE status = 'pending'");
259 $row = mysql_fetch_row($query);
260 $pending_total = $row[0];
261 if ($pending_total == null) {
262     $pending_total = 0;
263 }
264
265 $activity_q = mysql_query($conn,
266 "SELECT 'Production' AS type, pl.name AS item, pe.new_stock AS qty, 'x' AS direction, pe.entry_date AS date
267 FROM production_entries pe JOIN production_item pl ON pe.item_id = pl.id ORDER BY pe.entry_date DESC LIMIT 5)
268 UNION ALL
269 (SELECT 'Sale' AS type, pl.name AS item, s.qty AS qty, '-' AS direction, s.date AS date
270 FROM sales s JOIN production_item pl ON s.product_id = pl.id ORDER BY s.date DESC LIMIT 5)
271 ORDER BY date DESC LIMIT 8");
272
273 $alerts = [];
274
275 $result = mysql_query($conn, "SELECT material_name, current_stock, min_stock, unit FROM raw_material WHERE current_stock < min_stock AND current_stock > 0");
276 while ($r = mysql_fetch_assoc($result)) {
277     $alerts[] = ['icon' => '⬇️', 'msg' => ucfirst($r['material_name']) . ' stock below reorder level (' . $r['current_stock'] . ' vs ' . $r['unit'] . ' remaining)'];
278 }
279
280 $result = mysql_query($conn, "SELECT material_name FROM raw_material WHERE current_stock <= 0");
281 while ($r = mysql_fetch_assoc($result)) {
282     $alerts[] = ['icon' => '🔴', 'msg' => ucfirst($r['material_name']) . ' is out of stock'];
283 }
284
285 $result = mysql_query($conn, "SELECT name, current_stock FROM production_item WHERE current_stock < min_stock AND current_stock > 0");
286 while ($r = mysql_fetch_assoc($result)) {
287     $alerts[] = ['icon' => '🟡', 'msg' => $r['name'] . ' is running low (' . $r['current_stock'] . ' units)'];
288 }
289
290 $result = mysql_query($conn, "SELECT name FROM production_item WHERE current_stock <= 0");
291 while ($r = mysql_fetch_assoc($result)) {
292     $alerts[] = ['icon' => '🔴', 'msg' => $r['name'] . ' is out of stock'];
293 }
294
295 if ($pending_total > 0) {
296     $alerts[] = ['icon' => '💰', 'msg' => '€' . number_format($pending_total) . ' in pending payments outstanding'];
297 }
298
299 if (count($alerts) == 0) {
300     $alerts[] = ['icon' => '✅', 'msg' => 'All stock levels are healthy'];
301 }
302
303 $snapshot_rm = mysql_query($conn, "SELECT material_name AS item, 'Raw Material' AS category, current_stock AS available, unit, min_stock AS reorder FROM raw_material ORDER BY id");
304 $snapshot_fg = mysql_query($conn, "SELECT name AS item, 'Finished Goods' AS category, current_stock AS available, 'units' AS unit, min_stock AS reorder FROM production_item ORDER BY id");
305
306 function convert_money($amount) {
307     $amount = (int)$amount;
308     if ($amount >= 100000000) {
309         return '€' . number_format($amount / 100000000, 2) . ' Cn';
310     } else if ($amount >= 100000) {
311         return '€' . number_format($amount / 100000, 2) . ' L';
312     } else if ($amount >= 1000) {
313         return '€' . number_format($amount / 1000, 2) . ' K';
314     } else {
315         return '€' . number_format($amount);
316     }
317 }
318
319 function time_ago($date) {
320     $now = new DateTime();
321     $past = new DateTime($date);
322     $diff = $now->diff($past);
323 }
```

Production Management page :-

```
1 <?php
2 include '../config.php';
3 include 'login_check.php';
4 $page = 'production_management';
5 include 'access.php';
6 check_access($page);
7
8 function production_status($current, $sein) {
9     if ($current > $sein) {
10         return ['label' => 'Ready for sale', 'color' => '#34d399'];
11     } else if ($current <= 0) {
12         return ['label' => 'Out of Stock', 'color' => '#ff7f71'];
13     } else {
14         return ['label' => 'Low Stock', 'color' => '#ffb724'];
15     }
16 }
17
18 function get_raw_stock($conn) {
19     $result = mysqli_query($conn, "SELECT material_name, current_stock FROM raw_material");
20     $map = [];
21     while ($r = mysqli_fetch_assoc($result)) {
22         $map[$r['material_name']] = (int)$r['current_stock'];
23     }
24     return $map;
25 }
26
27 if ($SERVER['REQUEST_METHOD'] == 'POST' && isset($_POST['action']) && $_POST['action'] == 'add') {
28
29     $product_id = (int)$_POST['product_id'];
30     $produced_stock = (int)$_POST['produced_stock'];
31     $aggregate = (int)$_POST['Aggregate'];
32     $sand = (int)$_POST['sand'];
33     $cement = (int)$_POST['Cement'];
34     $red_sand = (int)$_POST['Red_sand'];
35
36     if ($aggregate == 0 && $sand == 0 && $cement == 0 && $red_sand == 0) {
37         header("Location: production_management.php?error=no-materials");
38         exit();
39     }
40
41     $raw = get_raw_stock($conn);
42
43     $not_enough = false;
44     if ($aggregate > ($raw['Aggregate'] ?? 0)) $not_enough = true;
45     if ($sand > ($raw['sand'] ?? 0)) $not_enough = true;
46     if ($cement > ($raw['Cement'] ?? 0)) $not_enough = true;
47     if ($red_sand > ($raw['Red Sand'] ?? 0)) $not_enough = true;
48
49     if ($not_enough) {
50         header("Location: production_management.php?error=insufficient-material");
51         exit();
52     }
53
54     $prod_stat = $conn->prepare("SELECT current_stock, min_stock FROM production_item WHERE id = ?");
55     $prod_stat->bind_param("i", $product_id);
56     $prod_stat->execute();
57     $prod = $prod_stat->get_result()->fetch_assoc();
58
59     if (!$prod) {
60         header("Location: production_management.php?error=invalid-product");
61         exit();
62     }
63
64     $available = (int)$prod['current_stock'] + $produced_stock;
65
66     if ($available > (int)$prod['min_stock']) {
67         $status = "Available stock";
68     } else {
69         $status = "Low Stock";
70     }
71
72     $stat = $conn->prepare("INSERT INTO production_entries (item_id, new_stock, Available, aggregate_used, sand_used, cement_used, red_sand_used, status) VALUES (?, ?, ?, ?, ?, ?, ?, ?)");
73     $stat->bind_param("iilliss", $product_id, $produced_stock, $available, $aggregate, $sand, $cement, $red_sand, $status);
74     $stat->execute();
75
76     $upd = $conn->prepare("UPDATE production_item SET current_stock = ? WHERE id = ?");
77     $upd->bind_param("ii", $available, $product_id);
78     $upd->execute();
79
80     $materials = [
81         'Cement' => $cement,
82         'Sand' => $sand,
83         'Aggregate' => $aggregate,
84         'Red Sand' => $red_sand,
85     ];
86
87     foreach ($materials as $name => $used) {
88         if ($used > 0) {
89             $r = $conn->prepare("UPDATE raw_material SET current_stock = current_stock - ? WHERE material_name = ?");
90             $r->bind_param("is", $used, $name);
91             $r->execute();
92         }
93     }
94
95     header("Location: production_management.php?success-added");
96     exit();
97 }
98
99 if ($SERVER['REQUEST_METHOD'] == 'POST' && isset($_POST['action']) && $_POST['action'] == 'update') {
100
101     $entry_id = (int)$_POST['entry_id'];
102     $new_stock = (int)$_POST['new_stock'];
103     $aggregate = (int)$_POST['Aggregate'];
104     $sand = (int)$_POST['sand'];
105     $cement = (int)$_POST['Cement'];
106     $red_sand = (int)$_POST['Red_sand'];
107
108     $old_stat = $conn->prepare("SELECT * FROM production_entries WHERE id = ?");
109     $old_stat->bind_param("i", $entry_id);
110     $old_stat->execute();
111     $old = $old_stat->get_result()->fetch_assoc();
112
113     if ($old) {
114
115         $item_id = (int)$old['item_id'];
116
117         $old_mats = [
118             'Aggregate' => (int)$old['aggregate_used'],
119             'Sand' => (int)$old['sand_used'],
120             'Cement' => (int)$old['cement_used'],
121             'Red Sand' => (int)$old['red_sand_used'],
122         ];
123
124         foreach ($old_mats as $name => $qty) {
125             $r = $conn->prepare("UPDATE raw_material SET current_stock = current_stock + ? WHERE material_name = ?");
126             $r->bind_param("is", $qty, $name);
127             $r->execute();
128         }
129
130         $old_new_stock = (int)$old['new_stock'];
131         $rev = $conn->prepare("UPDATE production_item SET current_stock = current_stock - ? WHERE id = ?");
132         $rev->bind_param("ii", $old_new_stock, $item_id);
133         $rev->execute();
134
135         $raw = get_raw_stock($conn);
```

```

136 $not_enough = false;
137 if ($aggregate > ($raw['Aggregate'] ?? 0)) $not_enough = true;
138 if ($sand > ($raw['Sand'] ?? 0)) $not_enough = true;
139 if ($cement > ($raw['Cement'] ?? 0)) $not_enough = true;
140 if ($red_sand > ($raw['Red Sand'] ?? 0)) $not_enough = true;
141
142 if ($not_enough) {
143     foreach ($old_mats as $name => $qty) {
144         $s = $conn->prepare("UPDATE raw_material SET current_stock = current_stock - ? WHERE material_name = ?");
145         $s->bind_param("is", $qty, $name);
146         $s->execute();
147     }
148     $rev2 = $conn->prepare("UPDATE production_item SET current_stock = current_stock + ? WHERE id = ?");
149     $rev2->bind_param("ii", $old_new_stock, $item_id);
150     $rev2->execute();
151     header("Location: production_management.php?error=insufficient_material");
152     exit();
153 }
154
155 $prod_stmt = $conn->prepare("SELECT current_stock, min_stock FROM production_item WHERE id = ?");
156 $prod_stmt->bind_param("i", $item_id);
157 $prod_stmt->execute();
158 $prod = $prod_stmt->get_result()->fetch_assoc();
159
160 $available = (int)$prod['current_stock'] + $new_stock;
161
162 if ($available > (int)$prod['min_stock']) {
163     $status = "Available stock";
164 } else {
165     $status = "Low Stock";
166 }
167
168 $upd_prod = $conn->prepare("UPDATE production_item SET current_stock = ? WHERE id = ?");
169 $upd_prod->bind_param("ii", $available, $item_id);
170 $upd_prod->execute();
171
172 $new_mats = [
173     'Aggregate' => $aggregate,
174     'Sand' => $sand,
175     'Cement' => $cement,
176     'Red Sand' => $red_sand,
177 ];
178
179 foreach ($new_mats as $name => $qty) {
180     if ($qty > 0) {
181         $s = $conn->prepare("UPDATE raw_material SET current_stock = current_stock + ? WHERE material_name = ?");
182         $s->bind_param("is", $qty, $name);
183         $s->execute();
184     }
185 }
186
187 $upd_entry = $conn->prepare("UPDATE production_entries SET new_stock=?, available=?, aggregate_used=?, sand_used=?, cement_used=?, red_sand_used=?, status=? WHERE id=?");
188 $upd_entry->bind_param("iiiiiii", $new_stock, $available, $aggregate, $sand, $cement, $red_sand, $status, $entry_id);
189 $upd_entry->execute();
190
191 header("Location: production_management.php?success=updated");
192 exit();
193
194 }
195
196 if (isset($_GET['delete_entry'])) {
197     $del_id = (int)$_GET['delete_entry'];
198
199     $old_stmt = $conn->prepare("SELECT * FROM production_entries WHERE id = ?");
200     $old_stmt->bind_param("i", $del_id);
201 }

```

Testing

What is Testing ?

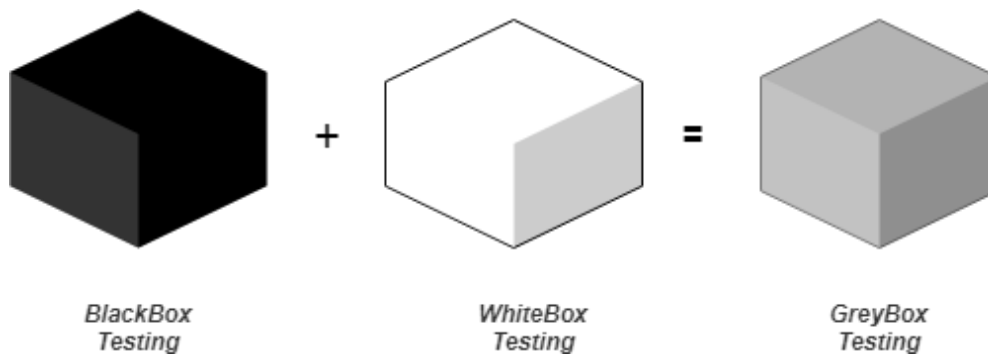
Testing is the process of evaluating and verifying that a software product or application.

Does what it is supposed to do.

Types of Testing

- Manual testing
- Automation testing

Manual Testing



➤ **White Box Testing:-**

- It is Software testing technique that
Involves testing the internal structure and working of a
software application.

➤ **Black Box Testing:-**

- A form of testing that is performed with no knowledge
of a systems
Internals.

➤ **Gray Box Testing:-**

- In Gray box testing, the tester is not required to design
test case.

Test Case

Login Form

Test case ID	Test Step	Input Data	Expected Result	Actual Result	Execution status
01	Enter Email and Password & click on Login button	Enter Valid Email & Password	User Should be go on Dashboard page	User Login Successfully	Successfully
02	Enter Email and Password & click on Login button	Enter invalid Email & Valid Password	Invalid email or password message will be display	User Not Login Successfully	Successfully
03	Enter Email and Password & click on Login button	Enter valid Email & invalid Password	Invalid email or password message will be display	User Not Login Successfully	Successfully
04	Enter Email and Password & click on Login button	Enter invalid Email & Password	Invalid email or password message will be display	User Not Login Successfully	Successfully
05	Enter any page with unauthorized access	Without email & password	Redirect to login page	Redirecting to login page	successfully

Pros and Cons For System

Advantage Of Project (Pros):-

- Reduce Manual Paper work and Error.
- Provide real time Stock Tracking.
- Save Time in Inventory Management.
- Easy Report Generation.
- Secure Role Based and login access Controls.

Limitations of Projects (Cons):-

- Requires Internet or System availability.
- Staff need basic training to use system.
- Risk of data loss without proper backup
- Dependence on computer system

Bibliography

Bibliography means take reference to prepare any project.

Here we are concern the book and websites which takes reference is as under.

Reference Websites:-

- <https://www.google.com>
- <https://www.youtube.com>
- <https://www.geeksforgeeks.org>